

MPHARM
(SEM. II) THEORY EXAMINATION 2025-26
ADVANCED PHARMACOGNOSY-II

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data: then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x10 = 20

- (a) Identify how to detect heavy metals.
- (b) Define reverse pharmacology.
- (c) Suggest some models for determination of antioxidant potentiality of herbal drugs.
- (d) Differentiate pharmacokinetic and pharmacodynamic studies
- (e) Define DNA fingerprinting.
- (f) What is bio prospecting?
- (g) Enlist the chemical constituents of *Curcuma longa*.
- (h) Write biological source and uses of amla and turmeric.
- (i) Discriminate ethno-botany and ethno-pharmacology
- (j) Define phyto-toxins with example

SECTION B

2. Attempt any two of the following:

2 x 10 = 20

- (a) Discuss the role of ethnopharmacology and reverse pharmacology in drug evaluation
- (b) Define adulteration. Discuss types of adulteration of herbal drugs and their sampling procedure in detail.
- (c) Describe the *in-vitro* and *in-vivo* evaluation techniques for anticancer and antidiabetic activities.

SECTION C

3. Attempt any five part of the following:

7 x 5 = 35

- (a) Explain DNA fingerprinting techniques for the identification of natural drugs
- (b) Explain the role of ethnobotany in herbal drug evaluation.
- (c) Discuss the analytical profile of *Andrographis paniculata*.
- (d) Narrate the analytical profile of *Boswellia serrata*.
- (e) Describe *in vitro* screening techniques for antimicrobial activity
- (f) How the toxicity studies can be done as per OECD guidelines
- (g) How will you determine microbial contamination in herbs and their formulations